

BUYAGA, Russian Federation

WMO: 310110

Lat: 59.6460N

Lon: 127.1220E

Elev: 278

StdP: 98.03

Time Zone: 9.00 (E09)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-50.5	-48.4	-52.1	0.0	-50.1	-50.0	0.0	-48.3	6.2	-21.5	5.5	-22.9	0.0	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.3	29.1	18.6	27.1	17.6	25.0	16.9	20.5	25.9	19.2	25.1	18.0	23.0	1.7	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.8	14.1	22.6	17.3	12.8	21.3	15.9	11.7	20.2	60.7	25.7	56.2	25.1	52.1	23.0	27.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.0	5.0	4.2	-52.3	32.0	2.3	2.0	-54.0	33.4	-55.4	34.6	-56.7	35.7	-58.4	37.2		
(5)			-52.6	22.4	2.3	2.3	-54.2	24.1	-55.6	25.5	-56.9	26.8	-58.6	28.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-8.0	-35.8	-30.1	-17.7	-4.9	6.2	14.4	17.2	14.1	5.2	-7.0	-25.3	-33.2		
	DBStd	19.99	8.20	7.56	7.73	6.28	4.33	3.64	3.52	3.79	4.04	6.55	8.25	8.92		
	HDD10.0	7071	1421	1123	860	446	131	7	1	9	149	527	1058	1340		
	HDD18.3	9639	1679	1357	1118	696	375	125	64	139	394	786	1308	1598		
	CDD10.0	518	0	0	0	0	14	139	223	136	6	0	0	0		
	CDD18.3	43	0	0	0	0	0	7	29	7	0	0	0	0		
	CDH23.3	824	0	0	0	0	13	232	411	168	1	0	0	0		
	CDH26.7	219	0	0	0	0	1	50	127	41	0	0	0	0		
(14) Wind	WSAvg	1.1	0.7	0.8	0.9	1.3	1.6	1.2	1.3	1.0	1.3	1.3	0.8	0.8		
(15) Precipitation	PrecAvg	597	21	17	21	32	55	71	96	91	77	55	35	25		
	PrecMax	760	42	38	58	75	109	135	188	220	179	104	75	56		
	PrecMin	452	3	6	5	13	12	10	30	16	16	28	14	10		
	PrecStd	81	10	7	10	14	20	25	42	45	33	19	15	9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-12.3	-7.4	5.9	14.9	25.3	30.7	32.6	30.1	20.8	9.5	1.5	-6.0		
		MCWB	-12.7	-8.5	1.3	7.0	14.2	18.8	19.9	19.1	13.3	5.2	0.3	-6.7		
	2%	DB	-16.0	-10.4	1.5	10.6	21.8	27.9	30.0	27.7	17.5	5.8	-4.7	-10.5		
		MCWB	-16.5	-11.2	-1.8	4.4	12.1	16.8	19.6	17.6	11.4	3.1	-5.4	-11.0		
	5%	DB	-19.5	-14.3	-0.9	8.5	18.4	26.1	27.6	25.1	15.0	3.5	-9.1	-15.6		
		MCWB	-19.9	-14.9	-3.1	2.8	10.2	16.4	19.0	17.0	10.3	1.4	-9.8	-16.0		
	10%	DB	-23.1	-17.4	-3.5	5.9	15.5	24.0	25.3	22.5	12.7	1.4	-13.0	-20.4		
		MCWB	-23.4	-17.9	-5.3	1.3	8.7	15.3	17.9	15.9	9.0	-0.2	-13.5	-20.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-12.7	-8.3	1.5	7.2	14.7	21.7	23.0	21.0	14.2	6.2	0.1	-6.7		
		MCDB	-12.3	-7.5	5.7	13.6	24.1	26.0	27.5	26.4	18.7	8.3	1.5	-6.0		
	2%	WB	-16.4	-11.1	-1.4	4.8	12.6	19.0	21.0	19.3	12.5	3.7	-5.1	-10.9		
		MCDB	-15.9	-10.3	1.0	10.0	19.7	25.1	25.9	25.0	16.6	5.6	-4.6	-10.4		
	5%	WB	-19.8	-14.8	-3.1	3.1	11.0	17.5	20.0	18.0	10.9	1.7	-9.8	-15.9		
		MCDB	-19.5	-14.3	-0.9	7.6	17.3	23.7	25.7	23.3	13.9	3.3	-9.1	-15.6		
	10%	WB	-23.4	-17.8	-5.3	1.6	9.5	16.1	18.9	16.6	9.4	-0.1	-13.5	-20.7		
		MCDB	-23.1	-17.4	-3.4	5.9	14.6	22.1	24.6	21.5	12.3	1.4	-13.0	-20.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.3	15.3	19.9	16.8	13.4	14.7	13.3	14.0	11.9	9.8	10.6	9.2		
		MCDBR	12.5	15.7	20.0	17.5	19.6	19.6	17.5	19.0	16.5	10.6	12.3	11.5		
		MCWBR	12.1	14.9	16.7	11.2	10.8	10.0	8.2	9.8	10.6	7.7	11.4	11.1		
	5% WB	MCDBR	12.5	15.7	19.0	16.2	17.4	16.7	15.2	15.6	15.4	9.7	12.2	11.6		
		MCWBR	12.2	15.0	16.0	11.0	10.3	9.9	8.6	8.3	10.5	7.5	11.5	11.1		
(40) Clear-Sky Solar Irradiance	taub		0.243	0.276	0.312	0.381	0.453	0.430	0.496	0.445	0.346	0.300	0.247	0.220		
	taud		1.984	2.034	2.031	1.961	2.028	2.231	2.042	2.163	2.413	2.271	2.016	1.921		
	Ebn at Noon		518	710	802	802	771	803	729	740	780	691	502	390		
	Edh at Noon		50	83	116	150	152	127	150	123	81	66	47	35		
(44) All-Sky Solar Radiation	RadAvg		0.57	1.63	3.36	5.14	5.11	5.73	5.17	4.06	2.57	1.34	0.74	0.33		
	RadStd		0.04	0.07	0.12	0.19	0.37	0.45	0.41	0.30	0.20	0.14	0.06	0.01		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page